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Set-8

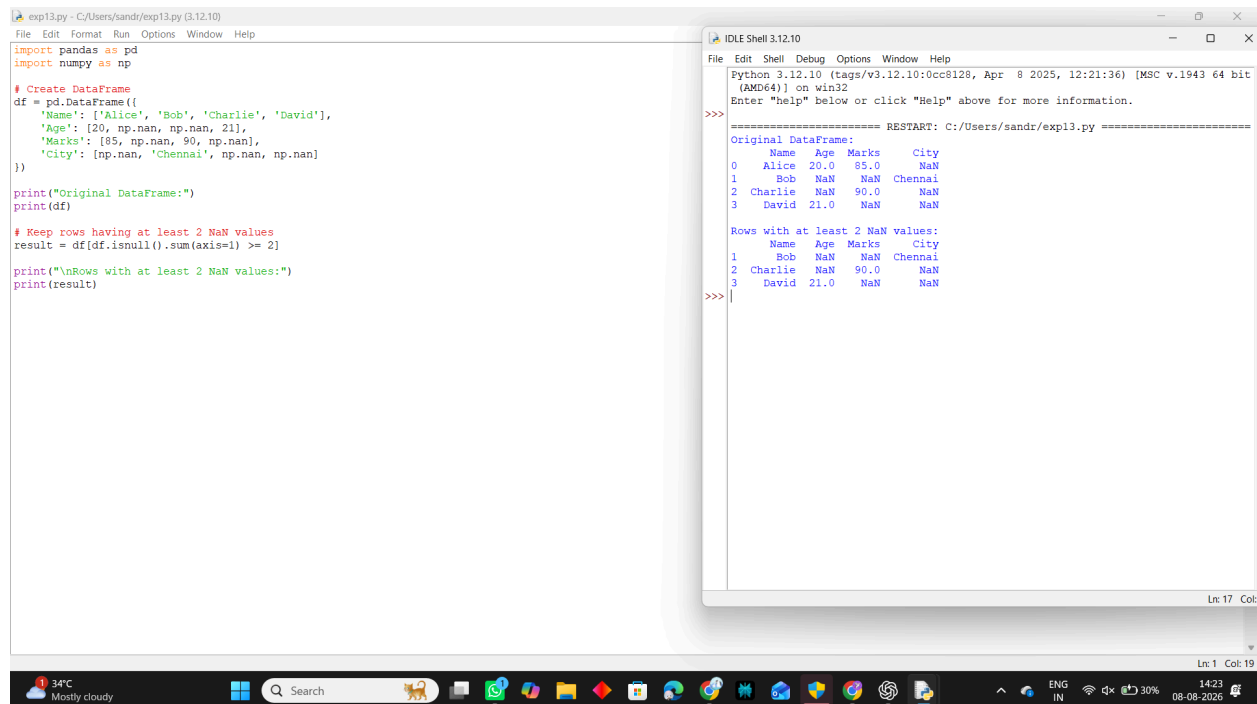
15. Write a Pandas program to keep the rows with at least 2 NaN values in a given DataFrame.

Aim

To write a Pandas program to keep the rows that contain **at least 2 NaN (missing) values** in a given DataFrame.

Procedure

1. Import the Pandas library and NumPy library.
2. Create a DataFrame containing some NaN values.
3. Use `isnull()` to identify missing values.
4. Use `sum(axis=1)` to count the number of NaN values in each row.
5. Select rows where the count of NaN values is **2 or more**.
6. Display the resulting DataFrame.



The screenshot shows a Python IDE with two windows. The left window displays the source code for a script named `exp13.py`. The code imports `pandas` and `numpy`, creates a DataFrame with columns 'Name', 'Age', 'Marks', and 'City', and then filters it to keep only rows with at least 2 NaN values. The right window shows the output of the script, which displays the original DataFrame and the resulting DataFrame after filtering.

```
exp13.py - C:/Users/sandr/exp13.py (3.12.10)
File Edit Format Run Options Window Help
import pandas as pd
import numpy as np

# Create DataFrame
df = pd.DataFrame({
    'Name': ['Alice', 'Bob', 'Charlie', 'David'],
    'Age': [20, np.nan, np.nan, 21],
    'Marks': [85, np.nan, 90, np.nan],
    'City': [np.nan, 'Chennai', np.nan, np.nan]
})

print("Original DataFrame:")
print(df)

# Keep rows having at least 2 NaN values
result = df[df.isnull().sum(axis=1) >= 2]

print("\nRows with at least 2 NaN values:")
print(result)
```

```
IDLE Shell 3.12.10
File Edit Shell Debug Options Window Help
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr 8 2025, 12:21:36) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Users/sandr/exp13.py =====
Original DataFrame:
   Name  Age  Marks  City
0  Alice  20.0   85.0   NaN
1   Bob   NaN   NaN  Chennai
2 Charlie  NaN   90.0   NaN
3  David  21.0   NaN   NaN

Rows with at least 2 NaN values:
   Name  Age  Marks  City
1   Bob   NaN   NaN  Chennai
2 Charlie  NaN   90.0   NaN
3  David  21.0   NaN   NaN

>>>
```

Result

Thus, the Pandas program was successfully executed to display the rows containing at least 2 missing (NaN) values.

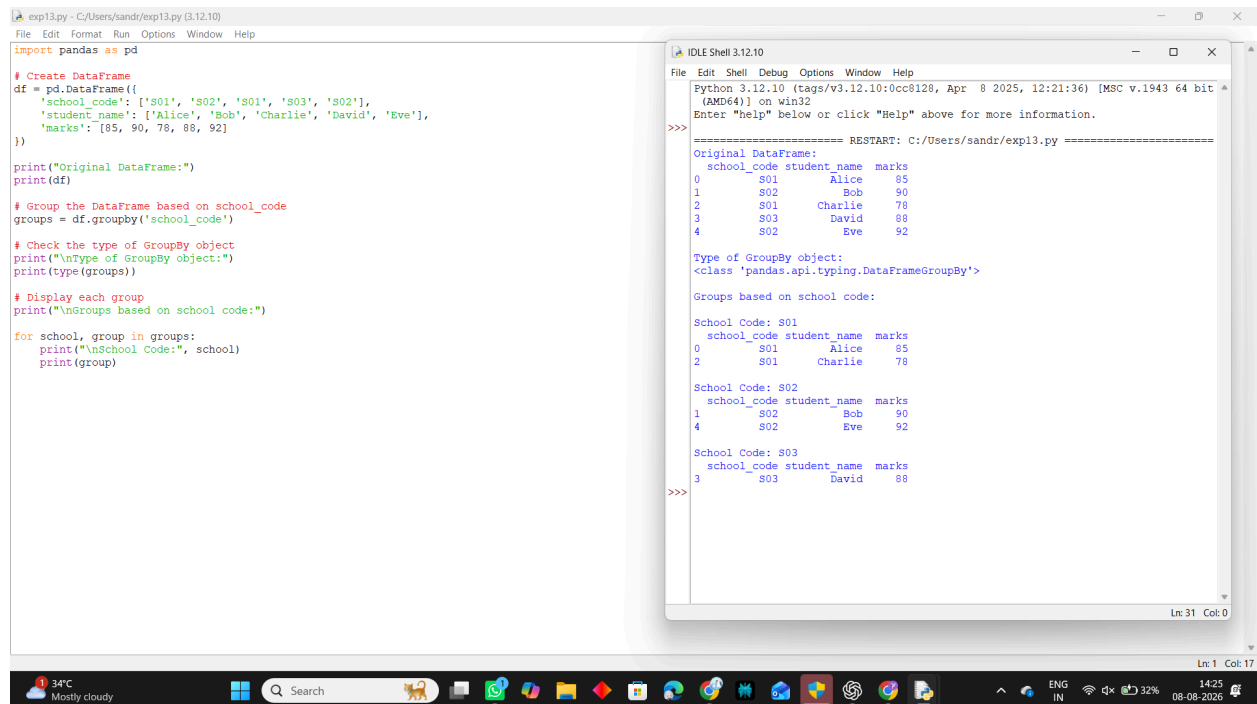
16. Write a Pandas program to split the following dataframe into groups based on school code. Also check the type of GroupBy object.

Aim

To write a Pandas program to split a given DataFrame into groups based on **school code** and check the type of the **GroupBy** object.

Procedure

1. Import the Pandas library.
2. Create a DataFrame containing student details and school codes.
3. Use **groupby()** to group the DataFrame based on the **school_code** column.
4. Check the type of the resulting GroupBy object using **type()**.
5. Display each group separately.



The screenshot shows a Python IDE with two windows. The left window displays the source code for a program that creates a DataFrame, groups it by 'school_code', checks the type of the resulting GroupBy object, and prints each group. The right window shows the output of the program, which includes the original DataFrame, the type of the GroupBy object, and the individual groups for each school code.

```
exp13.py - C:/Users/sandr/exp13.py (3.12.10)
File Edit Format Run Options Window Help
import pandas as pd

# Create DataFrame
df = pd.DataFrame({
    'school_code': ['S01', 'S02', 'S01', 'S03', 'S02'],
    'student_name': ['Alice', 'Bob', 'Charlie', 'David', 'Eve'],
    'marks': [85, 90, 78, 88, 92]
})

print("Original DataFrame:")
print(df)

# Group the DataFrame based on school_code
groups = df.groupby('school_code')

# Check the type of GroupBy object
print("\nType of GroupBy object:")
print(type(groups))

# Display each group
print("\nGroups based on school code:")

for school, group in groups:
    print("\nSchool Code:", school)
    print(group)
```

```
IDLE Shell 3.12.10
File Edit Shell Debug Options Window Help
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr 8 2025, 12:21:36) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Users/sandr/exp13.py =====
Original DataFrame:
  school_code student_name marks
0         S01         Alice    85
1         S02          Bob    90
2         S01        Charlie    78
3         S03         David    88
4         S02          Eve    92

Type of GroupBy object:
<class 'pandas.api.typing.DataFrameGroupBy'>

Groups based on school code:

School Code: S01
  school_code student_name marks
0         S01         Alice    85
2         S01        Charlie    78

School Code: S02
  school_code student_name marks
1         S02          Bob    90
4         S02          Eve    92

School Code: S03
  school_code student_name marks
3         S03         David    88

>>>
```

Result

Thus, the DataFrame was successfully **split into groups based on school code**, and the type of the GroupBy object was identified as **DataFrameGroupBy**.